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# Experiment: Iris Flower Classification using Naïve Bayes Classifier
Print("J THASLIMA NASREEN/192424152")

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

# Load Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# Display dataset information

print("Feature Names:")

print(iris.feature_names)

print("\nTarget Classes:")

print(iris.target_names)

# Split dataset into training and testing data

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Create Naïve Bayes classifier

model = GaussianNB()

# Train the classifier

model.fit(X_train, y_train)

# Predict test data

y_pred = model.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

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print("\nActual Values:")
print(y_test)
print("\nPredicted Values:")
print(y_pred)
print("\nAccuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")
# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
# Display confusion matrix
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=iris.target_names
)
disp.plot(cmap="Blues")
plt.title("Iris Flower Classification using Naïve Bayes")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.show()

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OUTPUT:

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Feature Names:

['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

Target Classes:

['setosa' 'versicolor' 'virginica']

Actual Values:

[0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]

Predicted Values:
[0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 2 1 0 2 0]

Accuracy: 0.9666666666666667
Accuracy Percentage: 96.66666666666667 %

Confusion Matrix:
[[10 0 0]
[0 9 1]
[0 0 10]]

